SOP-114 — Centrifugal Pump Suction Strainer Maintenance

Plant Site A · Rev 3 · Issued 14/08/2024 · Approved by Plant Engineering

1. Purpose

This procedure defines the method and the interval for inspecting and cleaning the temporary conical and permanent basket suction strainers fitted to the centrifugal process transfer pumps at Plant Site A. A plugged suction strainer starves the pump, causes cavitation and low discharge pressure trips, and is a leading cause of avoidable corrective work orders on the pump fleet.

2. Scope

Applies to the suction strainers fitted to the P-101 series process transfer pumps, namely Pump 101A, Pump 101B and Pump 101C, and to the basket strainers on the utilities cooling water pumps. The procedure covers routine cleaning, inspection of the strainer element, and the differential pressure surveillance that triggers unscheduled cleaning.

3. References

SOP-118 Equipment Isolation and Lock-Out Tag-Out.

SOP-101 Centrifugal Pump Startup and Shutdown Procedure.

Strainer manufacturer drawing showing element mesh size and gasket details.

4. Safety Precautions

The pump shall be stopped, and the strainer housing isolated, drained and depressurised per SOP-118 before the cover is unbolted. Strainer housings hold trapped pressure; crack the vent before the final fasteners.

Debris removed from strainers may include sharp metal fragments and process residue. Handle elements with cut-resistant gloves and dispose of debris to the designated bin.

Where the standby pump is placed in service to allow cleaning, complete the changeover per SOP-101 before isolating the pump to be worked.

5. Tools and Materials

Have available: strainer housing cover spanners, a new gasket of the specified material and size for the housing joint, cut resistant gloves, a soft bristle brush and low pressure wash water, a replacement strainer element from stores for exchange, a camera or work tablet for as-found photographs, a debris bin, and the work order for the pump tag.

Verify the pump tag against the work order at the machine. The three units of the P-101 set are identical in the field, and cleaning the wrong strainer leaves the fouled unit in service with a clean paper record.

6. Procedure

6.1 Confirm which pump of the set is to be worked and complete the changeover to the standby pump per SOP-101, verifying stable duty on the standby before isolating.

6.2 Isolate, drain and vent the strainer housing per SOP-118. Verify zero pressure at the vent before removing the cover fasteners.

6.3 Remove the cover and withdraw the strainer element. Photograph the as-found condition before any cleaning; the debris type and quantity is the primary record for interval review.

6.4 Estimate and record the percentage blockage of the element open area. Note the debris character: silt, scale flakes, gasket fragments, vegetation or polymer lumps each point to a different upstream source.

6.5 Compare the differential pressure history since the last cleaning against the 0.35 bar action limit, and note on the work order whether the limit was reached before the scheduled date. Early limit hits are the trigger for cleaning interval review.

6.6 Wash the element with low pressure water and a soft brush. Do not use wire brushes or high pressure jetting on fine mesh elements, as mesh distortion passes debris to the pump it is meant to protect.

6.7 Inspect the element for tears, distortion and corrosion. Hold the element against the light; any visible tear or opened mesh requires element replacement, not repair.

6.8 Where the debris volume is unusually high or the debris character has changed, inspect the suction line and the tank outlet screen for the source before returning the pump to duty, and report the finding to the shift supervisor.

6.9 Inspect the housing gasket face, fit a new gasket of the specified material, refit the element the correct way round per the flow arrow, and torque the cover fasteners evenly.

6.10 Return the strainer to service, venting the housing as it fills, and check for leakage at the cover joint at operating pressure.

6.11 Record the differential pressure across the strainer with the pump at normal duty. This reading is the clean baseline for surveillance until the next cleaning.

7. Cleaning Interval

Strainer elements on the P-101 series transfer pumps shall be removed, inspected and cleaned at monthly intervals. The monthly interval applies year round and is the basis of the preventive maintenance schedule for these pumps.

In addition to the scheduled monthly cleaning, an unscheduled cleaning shall be performed whenever the strainer differential pressure exceeds 0.35 bar at normal flow, or whenever pump suction pressure is observed trending down at constant tank level.

The cleaning interval shall not be extended without a documented review by the reliability engineer of the recorded blockage percentages over the preceding twelve months.

8. Acceptance Criteria

The task is accepted when the element is confirmed intact, the housing is leak free at operating pressure, the post-cleaning differential pressure is recorded as the new baseline, and the work order carries the blockage estimate and debris description for trend analysis.

9. Records

Record every cleaning, scheduled or unscheduled, as a work order against the pump tag with the blockage percentage and debris type. The reliability engineer reviews these records annually to confirm the cleaning interval remains appropriate. A rising blockage trend at a constant interval is the leading indicator that the interval requires revision.